

DeviceMesh

shape (2, 4), eight devices

axis 0: "data"

2 replicas

axis 1: "model"

each replica split
across 4 devices

LayoutMap

per variable, per dimension:
shard or replicate

```
graph TD; DM["DeviceMesh  
shape (2, 4), eight devices"] --> A0["axis 0: 'data'  
2 replicas"]; DM --> A1["axis 1: 'model'  
each replica split  
across 4 devices"]; A0 --> LM["LayoutMap  
per variable, per dimension:  
shard or replicate"]; A1 --> LM;
```